

INTERNATIONAL ISLAMIC UNIVERSITY ISLAMABAD

COURSE:-

Mobile Application Development
Course Code:- IT 332

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1. Project Overview & Objectives

This documentation details the design, architecture, and implementation of an interactive Sign-Up User Interface (UI) for an Android mobile application. Built as a core practical assignment for the Mobile Application Development course, the project demonstrates key Android engineering principles including XML layout design, Material Design components, Java activity lifecycle management, client-side input validation, and activity navigation using explicit Intents.

- **Material Design Form Components:** Outlined input text fields with rounded borders (12dp corner radius) and floating hint labels.
- **Password Masking & Visibility Toggle:** Integrated eye-toggle icons inside password fields using app:passwordToggleEnabled.
- **Keyboard Optimization:** Specific inputType configurations and keyboard action controls for seamless input flow.
- **Responsive Layout:** Encapsulated inside a ScrollView with fillViewport enabled to prevent content clipping.
- **Input Validation:** Java-based input verification checking for empty fields, proper email syntax, minimum password length, and matching confirm-password fields.
- **Activity Navigation:** Explicit Intent transition from MainActivity to HomeActivity, coupled with session termination via finish().

2. System Architecture & Project Structure

The project adheres to the standard Model-View-Controller (MVC) separation of concerns. Visual representations are defined entirely in declarative XML resource layouts, while user event handling, form parsing, and application control logic are encapsulated inside Java Activity classes.

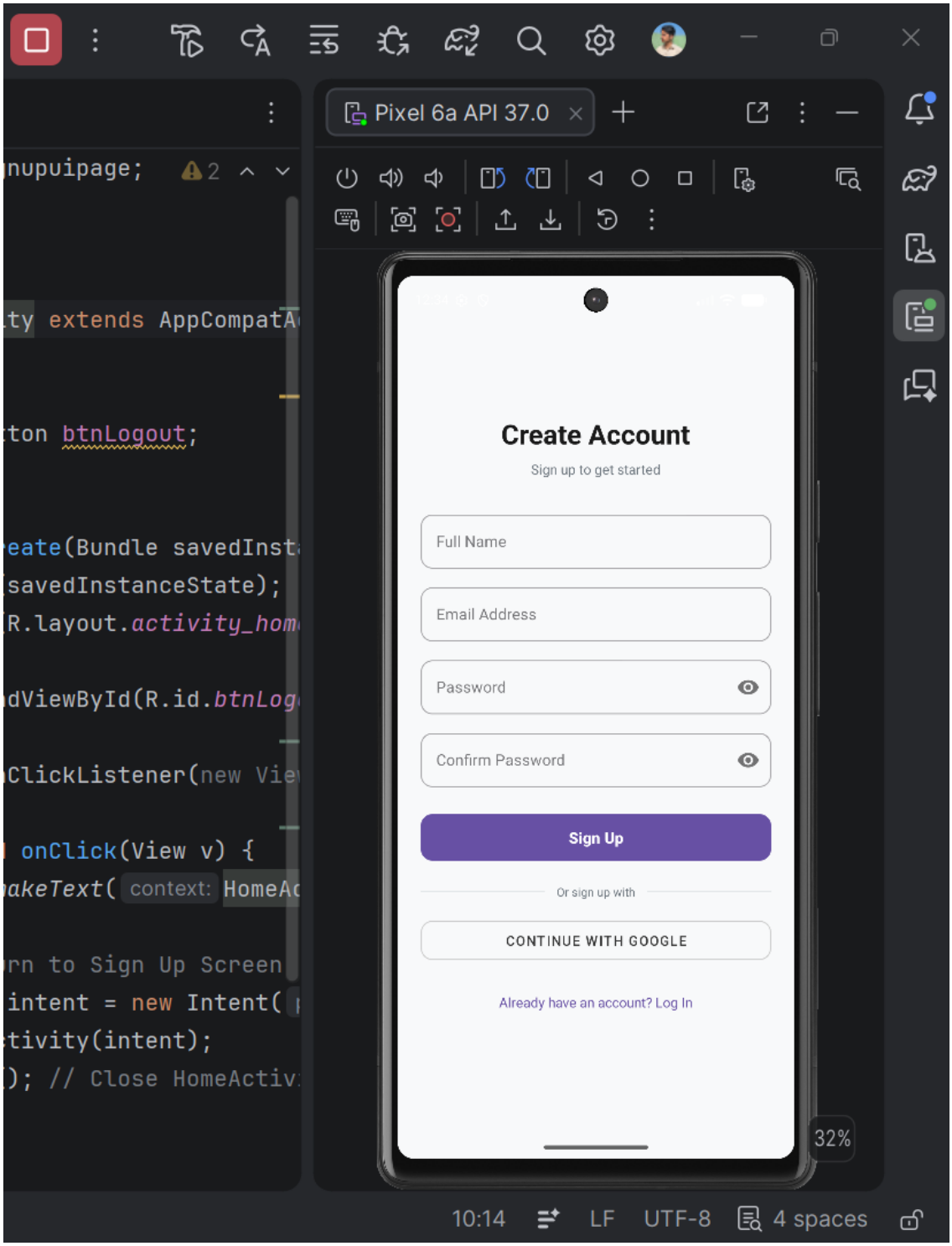
Directory Hierarchy

```
app/
├── manifests/
│   └── AndroidManifest.xml            # Central manifest registering activities
├── kotlin+java/
│   └── com.example.signupuipage/      # Java package containing Activity classes
│       ├── MainActivity.java          # Controller: Form Validation & Registration
│       └── HomeActivity.java          # Controller: Welcome Dashboard
└── res/
    └── layout/                        # XML Layout definitions
        ├── activity_main.xml          # View: Sign-Up UI Layout
        └── activity_home.xml          # View: Welcome Screen Layout
```

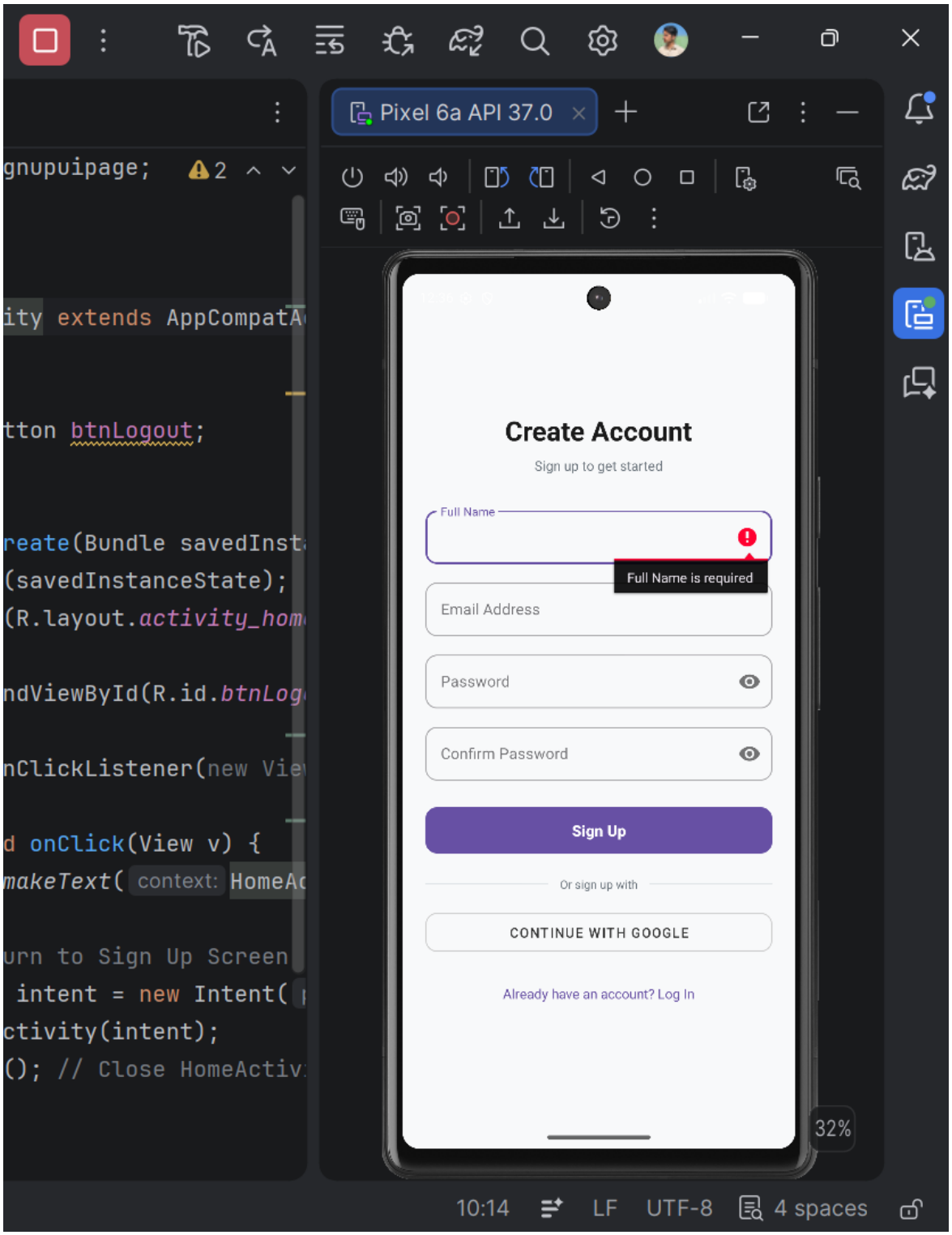
3. Application Screenshots

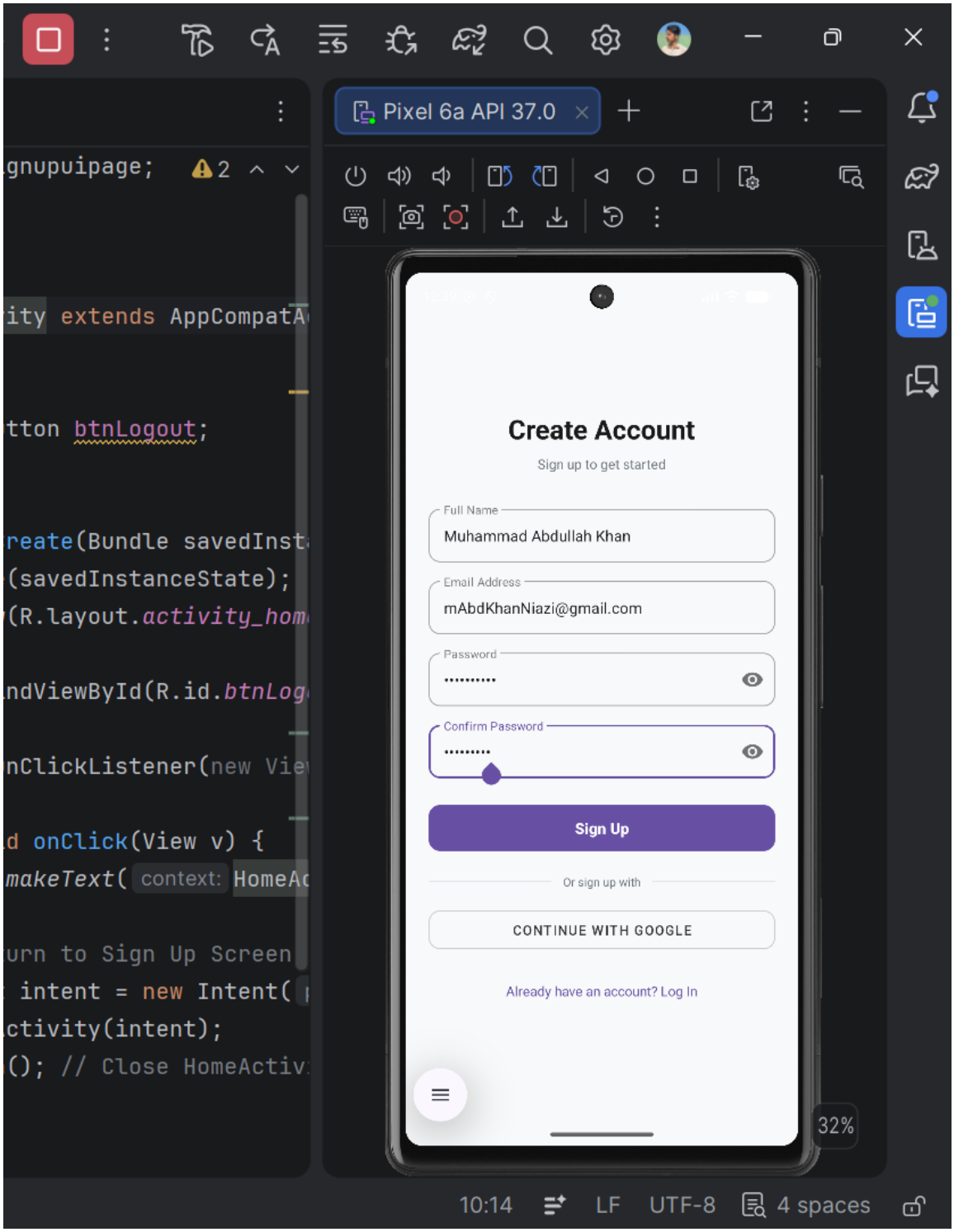
The following screenshots demonstrate the UI rendered on the Android Virtual Device (AVD), showcasing the Material Design components, layout structure, and validation error states.

3.1. Sign-Up Screen UI

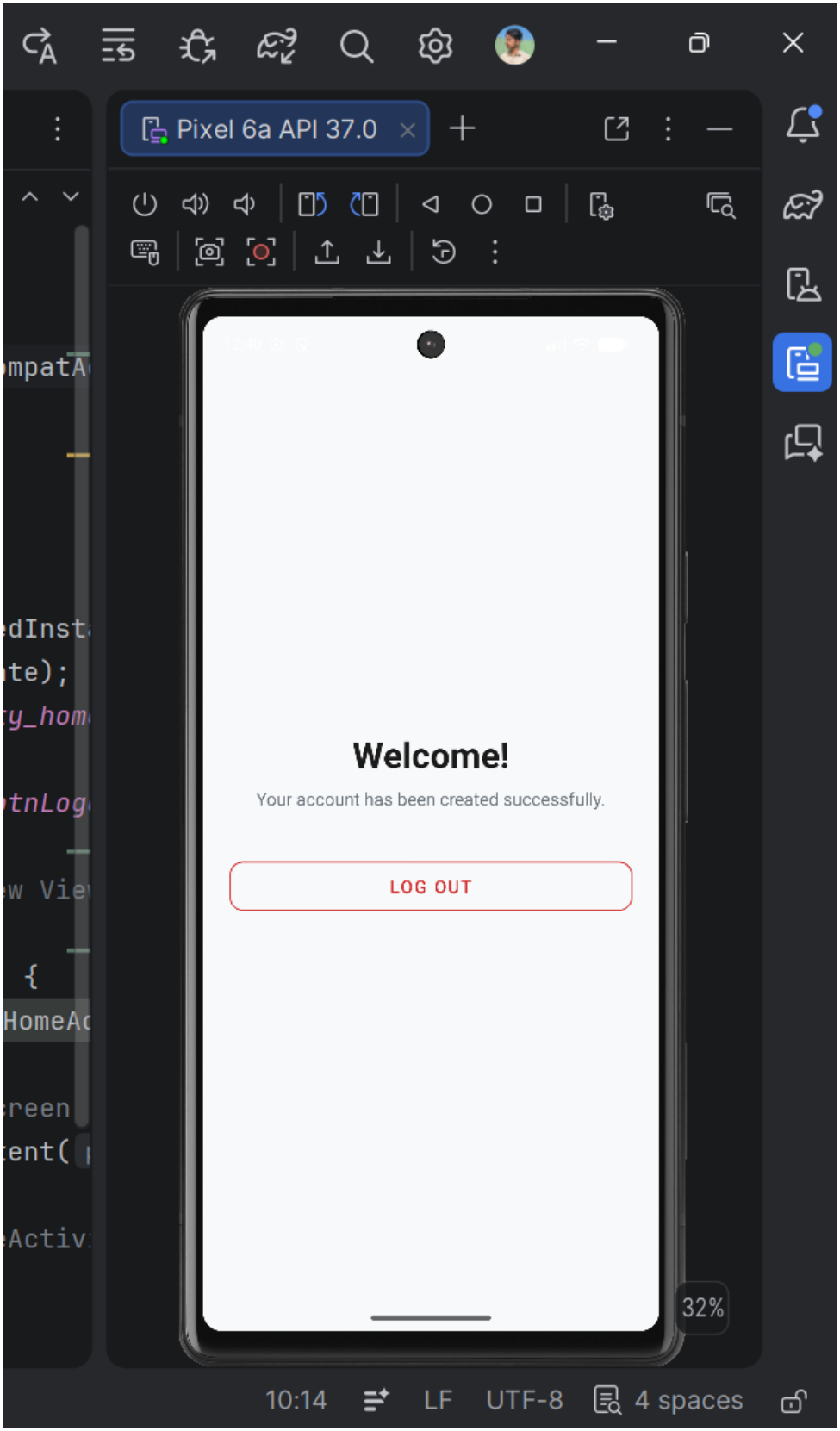


3.2. Form Validation Feedback




Pixel 6a API 37.0
Create Account
Sign up to get started
Full Name
Muhammad Abdullah Khan
Email Address
mAbdKhanNiazi@gmail.com
Password
Confirm Password
Sign Up
Or sign up with
CONTINUE WITH GOOGLE
Already have an account? Log In
32%
10:14
LF
UTF-8
4 spaces

3.3. Welcome Dashboard (Post-Login)



4. UI/UX Design System Specifications

UI Component	XML Element Type	Styling & Accessibility Features
Container	ScrollView	Allows smooth vertical scrolling when soft keyboard is shown; fillViewport ensures full-height stretch.
Main Layout	LinearLayout	Vertical orientation with 24dp padding for clean margin breathing space.
Text Inputs	TextInputLayout & TextInputEditText	OutlinedBox style, 12dp corner radius, floating hint labels, password toggles.
Primary Action	MaterialButton	Sign Up button with 56dp height, 12dp corner radius, bold 16sp text.
Secondary CTA	MaterialButton (Outlined)	Social sign-in button with subtle stroke outline (#CCCCCC) and gray text.

5. Validation & Exception Logic

Client-side validation executed in MainActivity.java guarantees data integrity before allowing navigation to the HomeActivity:

Field	Validation Condition	Error Response / UI Behavior
Full Name	name.isEmpty()	Displays 'Full Name is required', focuses etFullName field.
Email Address	email.isEmpty()	Displays 'Email is required', focuses etEmail field.
Email Syntax	!Patterns.EMAIL_ADDRESS.matcher	Displays 'Please enter a valid email address' for missing @ or domain.
Password	password.length() < 6	Requires minimum 6 characters for security compliance.
Confirm Password	!password.equals(confirmPassword)	Displays 'Passwords do not match', prevents form submission.

6. Testing & Troubleshooting Guide

Common AVD environment issues encountered during development:

Emulator Process Lock Issue

Symptom: Android Studio reports 'Pixel 6a is already running'.

Resolution: Terminate 'qemu-system-x86_64.exe' via Task Manager, perform a 'Cold Boot Now' from the Device Manager, or delete '.lock' files inside the AVD configuration directory.

Duplicate Activity Manifest Error

Symptom: Build fails due to duplicate HomeActivity declarations.

Resolution: Ensure each <activity> is declared exactly once inside AndroidManifest.xml.

7. Future Scope & Recommendations

The current implementation provides a robust baseline for client-side user onboarding. To elevate this project to production readiness, the following enhancements are recommended for future iterations:

- **Firebase Authentication Integration:** Replace the current UI-only Google sign-in button and local Toast validations with a live backend using Firebase Auth and Google Identity Services SDK.
- **User Session Persistence:** Utilize Android's EncryptedSharedPreferences or a Room Database to maintain user sessions so users do not need to log in repeatedly after closing the application.
- **Data Passing via Intents:** Enhance the navigation flow by passing the verified 'Full Name' string from MainActivity to HomeActivity via Intent extras (e.g., intent.putExtra()), updating the Dashboard to read 'Welcome, [Name]!'.
- **Motion & Transitions:** Introduce Lottie vector animations on the success screen and Shared Element Transitions between activities for a smoother, modern UX feel.